Supplementary Material
SP-BART: A Semi-parametric BART for estimating heterogeneous treatment effects

Yusufhan Balci, Mark van de Wiel and Jeroen Hoogland

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This supplement provides additional simulation and application results and computational details.

S1 Orthogonalisation of the Treatment Contrast

The standard fit uses centered treatment coding, $z - 0.5$. In the orthogonalised fit, we use $z - \hat{\pi}(x)$ instead. We estimate $\hat{\pi}(x)$ once with a Bayesian logistic regression including main effects and pairwise interactions, and hold it fixed across chains. Both fits use the same prognostic forest specification and prior. Residualising treatment changes the conditional outcome function represented by the prognostic component.

In this comparison, orthogonalising the treatment contrast increases CATE RMSE in every design, by 12.6 to 37.2% (Table S1). Under a linear μ, the increase averages 17.1%, with wider intervals and similar coverage. Under a non-linear μ, the increase averages 29.4% and coverage falls despite wider intervals. The last two columns of Table S1 use the same interval-based top-quartile rule as in the main simulations; sensitivity falls in both panels.

The standard-arm values come from this separate experiment and differ from those in main-text Table 1. There are 200 replicates at each sample size.

S2 Homogeneous Simulation Results

Table S2 presents the results for the homogeneous simulation scenario, where $\tau(x) = 3$ and all non-intercept treatment-effect coefficients are zero. The results show that SP-BART and BCF perform similarly and can shrink back towards homogeneity.

S3 Coverage-matched Individual CATE Inference on IHDP

This calibration is possible because the individual treatment effects are known in the semi-synthetic benchmark.

Table S1: Orthogonalised versus standard SP-BART under both prognostic forms, with 200 replicates per setting. Change is the percentage increase in CATE RMSE. Sensitivity uses the same interval-based top-quartile rule as in the main simulations.

	CATE RMSE			Coverage		CI width		Top-25% sensitivity	
N	standard	residualised	change	standard	residualised	standard	residualised	standard	residualised
Linear μ									
250	0.639	0.800	+25.1%	0.942	0.957	2.415	3.253	0.491	0.376
500	0.381	0.454	+19.1%	0.959	0.965	1.619	1.945	0.621	0.551
750	0.312	0.355	+13.5%	0.963	0.966	1.307	1.512	0.689	0.637
1000	0.260	0.304	+16.7%	0.965	0.966	1.116	1.287	0.732	0.686
1500	0.205	0.237	+15.8%	0.969	0.968	0.900	1.021	0.776	0.743
3000	0.143	0.161	+12.6%	0.967	0.968	0.623	0.697	0.843	0.823
Non-linear μ									
250	0.578	0.737	+27.5%	0.957	0.955	2.318	2.970	0.465	0.402
500	0.350	0.481	+37.2%	0.968	0.956	1.532	1.908	0.612	0.558
750	0.266	0.346	+30.1%	0.977	0.971	1.219	1.477	0.676	0.623
1000	0.223	0.295	+32.4%	0.978	0.969	1.037	1.254	0.717	0.659
1500	0.176	0.223	+26.5%	0.978	0.973	0.828	0.989	0.776	0.737
3000	0.123	0.152	+23.0%	0.979	0.976	0.574	0.677	0.841	0.812

Table S2: Performance Comparison in the **Homogeneous Scenario** (Constant CATE, $\tau = 3$). Results are shown for models with Homogeneous Treatment Effects, comparing Linear and Non-linear prognostic functions (μ).

		BCF						**SP-BART**					
$\mu(x)$ Form	N	ATE			CATE			ATE			CATE		
		RMSE	Cov	Len	RMSE	Cov	Len	RMSE	Cov	Len	RMSE	Cov	Len
Linear	250	0.326	0.82	0.852	0.494	0.96	2.039	0.348	0.80	0.859	0.590	0.94	2.268
	500	0.193	0.84	0.559	0.337	0.97	1.445	0.198	0.83	0.557	0.347	0.96	1.501
	750	0.150	0.88	0.442	0.275	0.98	1.179	0.153	0.85	0.441	0.284	0.96	1.216
	1000	0.136	0.81	0.377	0.243	0.98	1.018	0.137	0.81	0.376	0.238	0.97	1.036
	1500	0.100	0.86	0.303	0.201	0.98	0.827	0.101	0.84	0.303	0.187	0.97	0.836
	3000	0.069	0.89	0.210	0.142	0.97	0.569	0.068	0.89	0.211	0.132	0.96	0.576
Non-linear	250	0.247	0.93	0.825	0.470	0.97	1.934	0.250	0.92	0.810	0.549	0.95	2.188
	500	0.159	0.91	0.535	0.314	0.97	1.303	0.162	0.92	0.530	0.331	0.96	1.445
	750	0.109	0.94	0.424	0.235	0.99	1.068	0.109	0.96	0.422	0.248	0.98	1.150
	1000	0.089	0.95	0.363	0.207	0.98	0.938	0.090	0.95	0.360	0.205	0.98	0.981
	1500	0.080	0.93	0.291	0.175	0.98	0.761	0.080	0.91	0.288	0.166	0.97	0.782
	3000	0.051	0.95	0.201	0.124	0.99	0.530	0.049	0.96	0.200	0.115	0.98	0.545

The original 95% credible intervals of the two Bayesian methods do not have the same coverage of the true τ_i on IHDP, so their exclusion rates are not directly comparable. Let m_i be the posterior mean, and $q_{i,0.025}$ and $q_{i,0.975}$ the posterior percentiles that form the central 95% credible interval. We rescale this interval as

$$I_i(c) = [m_i + c\{q_{i,0.025} - m_i\},\ m_i + c\{q_{i,0.975} - m_i\}]\,.$$

The scale factor $c > 1$ widens the interval. For each held-out replicate and each model, we choose the smallest c attaining at least 95% empirical coverage across the other 99 data sets, and then evaluate it on the held-out replicate. The resulting intervals are empirically calibrated CATE intervals, not tolerance intervals or necessarily 95% posterior credible intervals. The shared covariates and treatment assignments mean this evaluates calibration across IHDP response surfaces rather than across new patient samples. Table S3 reports the comparison; held-out mean coverage is 0.950 for both methods. The exclusion rates are close.

Table S3: Coverage-matched individual CATE inference on IHDP. Each model's intervals are rescaled towards a common empirical coverage target of 95%. The scale factor is chosen from the other 99 data sets and then applied to the held-out one; held-out coverage need not equal the target exactly. Values are mean (standard error) over 100 replicates. The scale factor c multiplies the distances from the posterior mean to the 2.5th and 97.5th posterior percentiles; $c > 1$ widens the interval. "Correct exclusion" is the fraction of children whose interval excludes zero in the direction of the true effect.

Method	c	coverage	excludes 0	correct exclusion	Type S	width
SP-BART	1.105 (0.000)	0.950 (0.004)	0.876 (0.008)	0.875 (0.008)	0.001 (0.000)	5.56 (0.61)
BCF	1.186 (0.000)	0.950 (0.003)	0.878 (0.010)	0.876 (0.010)	0.002 (0.000)	5.39 (0.76)

S4 Detailed Analysis: The CANVAS Program

S4.1 Out-of-Sample Performance

We compare SP-BART with BCF by using an analytic sample of 5018 individuals from CANVAS-R for training and 3004 individuals from CANVAS for validation. These are the numbers included in this analysis, rather than the original trial enrolments. To study the effect of sample size, we drew 20 treatment-stratified samples without replacement for $N \in \{500, 1000, 1500, 2000, 3000, 4000\}$ from CANVAS-R and trained both models on each sample. We also fit both models once on the full CANVAS-R sample. Both models include the treatment proportion in the full CANVAS-R training pool as the same constant propensity score in the prognostic component. We evaluate agreement between the models using the correlation between their out-of-sample CATEs, and we evaluate ranking using the uplift curve (Rzepakowski and Jaroszewicz, 2010). Because the outcome is follow-up minus baseline HbA1c, a more negative CATE indicates greater benefit. The uplift curve ranks individuals from high to low predicted benefit, $-\hat{\tau}(x)$, and plots the cumulative gain as

an increasing portion of the population is treated. This evaluation relies on randomization; without exchangeability, the empirical treatment effect cannot be compared directly with the model output. Figure S1 and Table S4 show the results for both methods at the different sample sizes.

The comparison of the individual CATEs shows that the two models already agree reasonably well at small N, and that this agreement increases with sample size (Table S4 and Figure S1). At $N = 1000$ the average correlation of the out-of-sample CATEs is $r = 0.69$; at $N = 4000$ it is $r = 0.81$, and $r = 0.84$ when both models are trained on the whole CANVAS-R sample. Spearman correlations are similar. This indicates similar estimated treatment effect patterns, although agreement between models does not establish the true functional form. At smaller sample sizes, both models often shrink the treatment effect back towards homogeneity, BCF more strongly.

The uplift curves in Figure S1 are comparable for both models and remain close to the diagonal representing random targeting. This limits the evidence for improved targeting despite agreement between the models. The normalized net areas under the uplift curve in Table S4 are similar as well. Thus, even when the CATE magnitudes differ at lower sample sizes, the models rank patients similarly. SP-BART also has smaller RMSEs for the out-of-sample outcome (Y) at every training size in Table S4. In this evaluation, SP-BART therefore provides comparable ranking while also allowing direct variable selection, which we examine in the next section.

To standardize the area under the uplift curve, write $U(f)$ for cumulative uplift when the fraction f of the validation sample is selected. With $n_v = 3004$, we report

$$\frac{\text{NetAUUC}}{n_v} = \frac{1}{n_v}\left\{\int_0^1 U(f)\,df - \frac{U(1)}{2}\right\}.$$

We subtract the area under the straight line for random targeting and divide by the validation-sample size. Positive values indicate greater cumulative HbA1c reduction than random targeting. Table S4 reports this normalized measure.

At $N = 500$, the outcome-RMSE standard deviation is much lower for SP-BART than for BCF (0.016 versus 0.119). We interpret this difference as reflecting the greater sample-to-sample variation induced by BCF's tree-based treatment-effect structure.

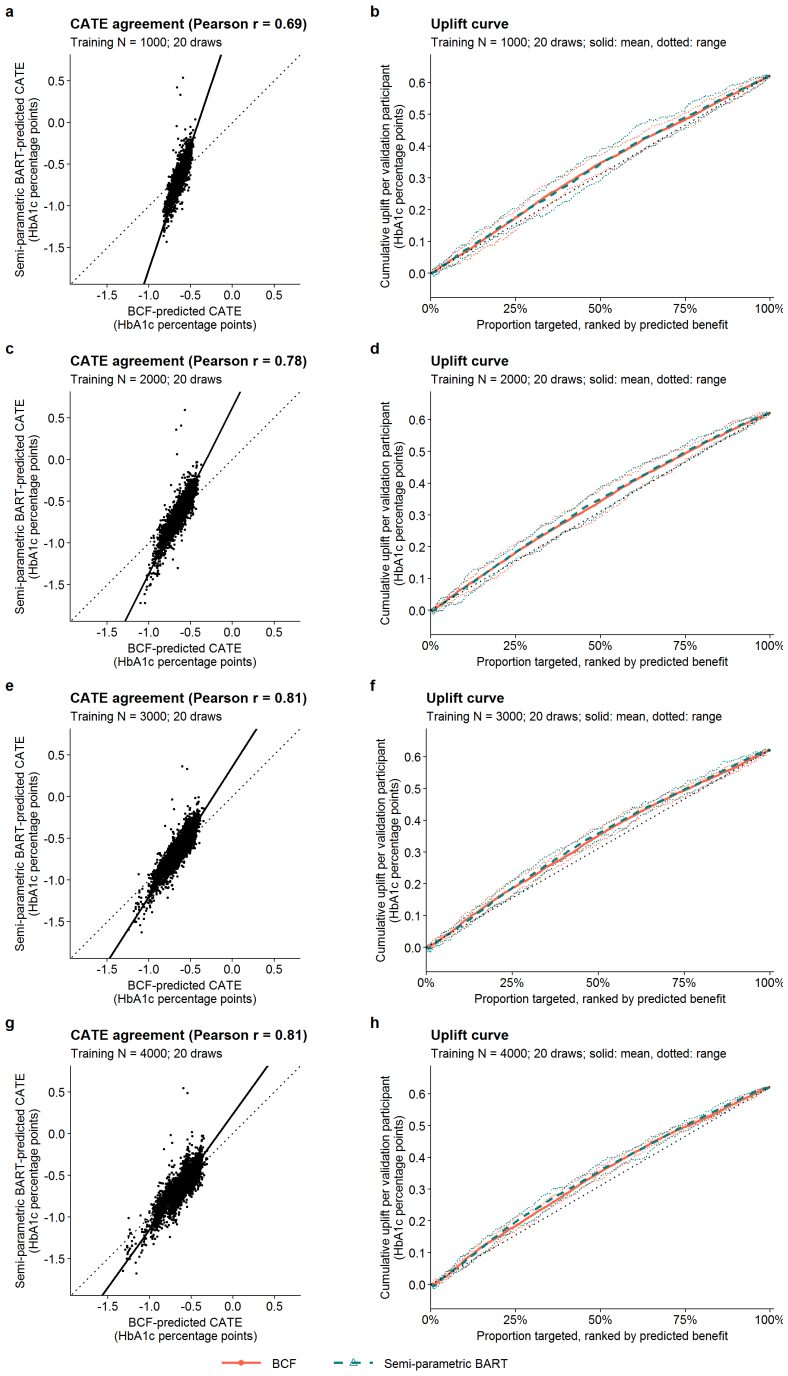


Figure S1: Out-of-sample CATE agreement (left) and uplift curves (right) for SP-BART and BCF at $N = 1000$, 2000, 3000, and 4000 (mean over 20 draws). The orthogonal fit is a total least-squares line between the two models' predictions, minimizing perpendicular distances; it is unrelated to treatment residualisation in Section S1. Dotted min–max lines show the pointwise range across the 20 training draws. The dotted diagonal marks equality in the agreement panels and random targeting in the uplift panels.

Table S4: Out-of-sample performance on CANVAS. Values are mean ± standard deviation over 20 training draws, except the last row, which is a single fit on the full CANVAS-R sample. RMSE is for the outcome Y. Correlation and Spearman are between the two models' out-of-sample CATEs. Net AUUC subtracts the random-targeting area and is divided by the 3004 validation participants; its units are HbA1c change.

		Outcome RMSE		CATE agreement		Net AUUC/n_v	
N	Draws	BCF	SP-BART	r	Spearman	BCF	SP-BART
500	20	1.009 ± 0.119	0.947 ± 0.016	0.702 ± 0.137	0.759 ± 0.088	0.009 ± 0.013	0.012 ± 0.015
1000	20	0.974 ± 0.078	0.926 ± 0.009	0.692 ± 0.130	0.727 ± 0.108	0.019 ± 0.010	0.021 ± 0.014
1500	20	0.952 ± 0.057	0.918 ± 0.004	0.741 ± 0.102	0.778 ± 0.090	0.020 ± 0.009	0.025 ± 0.008
2000	20	0.979 ± 0.044	0.916 ± 0.005	0.778 ± 0.088	0.803 ± 0.074	0.023 ± 0.009	0.026 ± 0.010
3000	20	0.970 ± 0.058	0.913 ± 0.002	0.811 ± 0.069	0.828 ± 0.052	0.025 ± 0.004	0.028 ± 0.006
4000	20	0.944 ± 0.024	0.910 ± 0.002	0.813 ± 0.033	0.823 ± 0.031	0.027 ± 0.003	0.031 ± 0.005
5018	1	0.950	0.909	0.843	0.851	0.029	0.035

S4.2 Interpretability and Variable Selection in Practice

The out-of-sample evaluations show comparable ranking for the two models. SP-BART also allows structural interpretation. We therefore applied the Shapley value decomposition to the CANVAS-R data to describe the covariate contributions to treatment effect heterogeneity. The Shapley summaries and the HbA1c and age coefficient posteriors are shown in the main text (Figures 2 and 3). Table S5 defines the abbreviations.

Table S5: Baseline covariates in the CANVAS analysis and figure labels.

Label	Covariate
BLBMI	Body mass index
AGE	Age
HHFFL	History of heart failure
BLSBP	Systolic blood pressure
SEX	Sex
DIADUR	Diabetes duration
HCEREBFL	History of cerebrovascular disease
BLA1C	HbA1c
BLGFR	Estimated glomerular filtration rate
INSULBL	Insulin use
GLP1BL	GLP-1 receptor agonist use
BLACR	Urinary albumin-to-creatinine ratio
RACE	Recorded race, represented by sum contrasts RACE1–RACE5
METFORBL	Metformin use
RANGIBL	Use of agents acting on the renin–angiotensin system

For RACE, the signed column contributions are summed within each individual before taking absolute values. RACE, baseline HbA1c, and age have the largest contributions, predominantly through main effects. The recorded race categories are White; Black or

African American; Asian; American Indian or Alaska Native; Native Hawaiian or Other Pacific Islander; and Other/Multiple. We combine Other and Multiple, and treat Unknown and Not Reported as missing before complete-case filtering.

To assess each covariate, we compare the full fitted CATE, including its interactions, at selected values while averaging over the empirical distribution of the other covariates. For continuous covariates, we compare the 90th and 10th percentiles; for binary covariates, the two levels; and for RACE, every pair of categories. For posterior contrast draws $d_\ell^{(m)}$, with posterior mean $\bar{d}_\ell$ and standard deviation s_ℓ, we take the 95th percentile k of $\max_\ell |(d_\ell^{(m)} - \bar{d}_\ell)/s_\ell|$. The intervals $\bar{d}_\ell \pm k s_\ell$ are simultaneous across all assessed contrasts. A covariate is flagged when at least one of its intervals excludes zero.

Figure S2 shows the simultaneous intervals for all assessed contrasts.

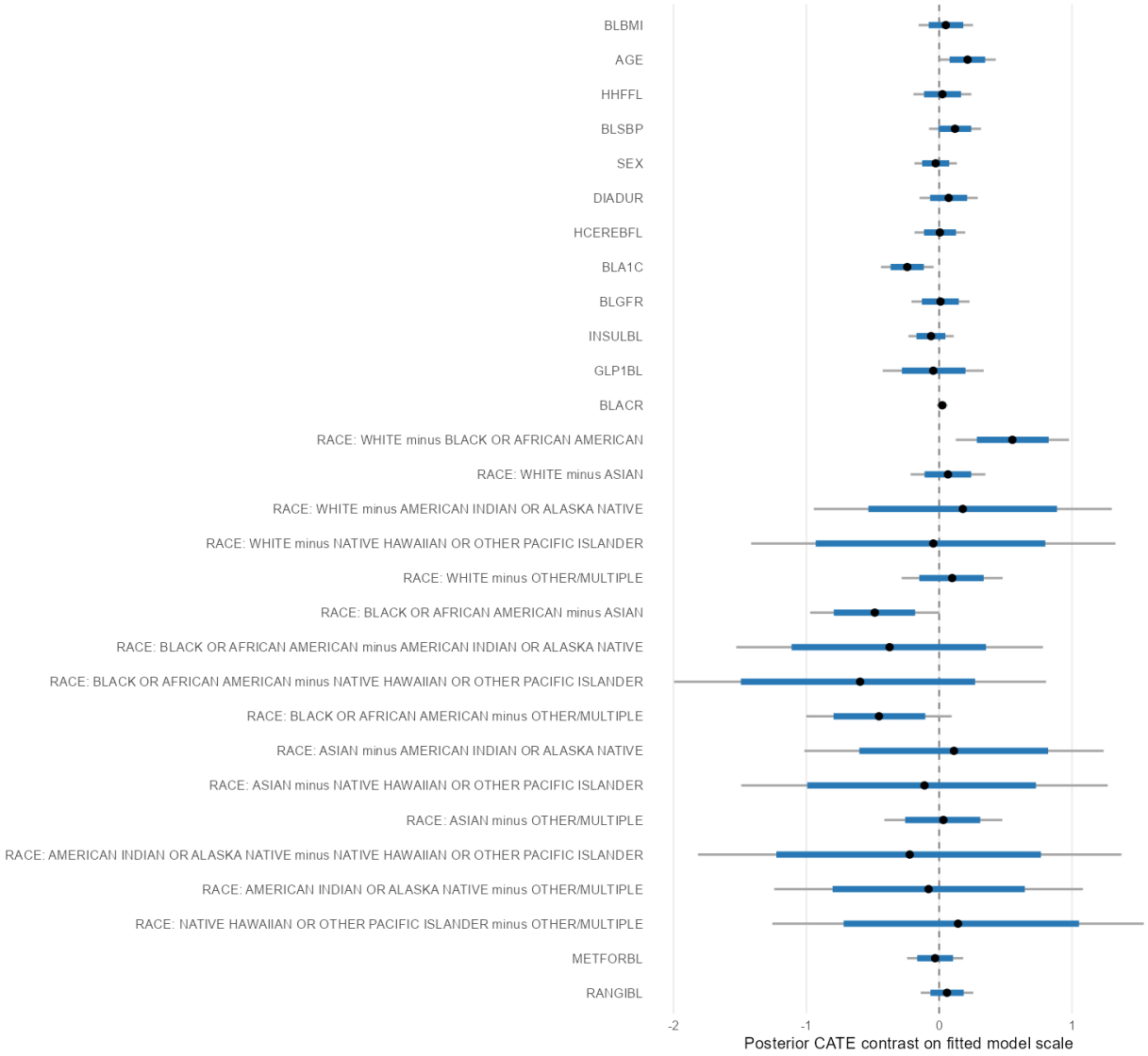


Figure S2: Full fitted CATE contrasts in CANVAS-R, including interactions and averaged over the other covariates. Continuous covariates compare the 90th minus 10th percentile; binary covariates compare the higher-coded minus lower-coded level; RACE compares the first named minus second named category. Points are posterior means; blue and grey intervals are pointwise and simultaneous 95% credible intervals, respectively. Contrasts are on the fitted model scale.

S5 Ranking Performance in Simulation

Tables S6 and S7 display interval-based selection performance under the Heavy Saturation and Mild Saturation scenarios, respectively.

Two patterns hold across both scenarios. The advantage of SP-BART is concentrated at the sharp thresholds. Under heavy saturation it attains both higher sensitivity and higher precision than BCF at the top 10% threshold, recovering on average 33.8% of the true top decile against 3.6% for BCF, at a mean precision of 0.991 against 0.743. At the top 50% threshold under heavy saturation, mean sensitivities are similar (0.725 for BCF against 0.722), while their mean precisions are close (0.997 against 0.996). The advantage under this selection rule is therefore concentrated at the more selective threshold.

Table S6 shows that under heavy saturation the precision of SP-BART stays at or above 0.957 at every sample size and threshold, while BCF's top 10% precision ranges from 0.468 to 0.920 and is weakest when $\mu(x)$ is non-linear. The selection rule can therefore identify high-benefit individuals in this monotone simulation, even though the CATE magnitudes are biased and coverage drops (Table 2). Under mild saturation (Table S7) the same ordering shows up from $N \geq 1000$ onwards, but at $N = 250$ neither model is dependable at the top decile: precision falls to 0.615 for SP-BART under a linear $\mu(x)$ and to 0.230 for BCF under a non-linear one. At small sample sizes, precision at the sharpest threshold is not stable. These ranking results are for a monotone treatment function.

Table S6: Interval-based selection performance: heavy saturation scenario. A patient is flagged if the lower bound of the 95% credible interval exceeds the threshold. Precision is averaged only over replicates in which the model flagged at least one patient, so it should be read together with sensitivity.

$\mu(x)$ Form	N	**BCF**						**SP-BART**					
		Top 10%		Top 25%		Top 50%		Top 10%		Top 25%		Top 50%	
		Sens	Prec	Sens	Prec	Sens	Prec	Sens	Prec	Sens	Prec	Sens	Prec
Linear	250	0.060	0.797	0.095	0.961	0.510	0.994	0.167	0.965	0.159	0.994	0.512	0.987
	500	0.060	0.870	0.147	0.993	0.666	0.999	0.251	0.992	0.237	1.000	0.675	0.996
	750	0.068	0.883	0.190	0.990	0.726	0.999	0.315	0.996	0.282	1.000	0.729	0.997
	1000	0.064	0.907	0.219	0.995	0.760	0.999	0.356	0.999	0.316	1.000	0.774	0.998
	1500	0.056	0.897	0.297	0.994	0.807	0.998	0.410	1.000	0.356	1.000	0.810	0.998
	3000	0.047	0.920	0.519	0.994	0.856	0.999	0.515	1.000	0.422	1.000	0.866	0.999
Non-linear	250	0.017	0.468	0.043	0.852	0.505	0.992	0.172	0.957	0.162	0.986	0.517	0.990
	500	0.015	0.560	0.092	0.927	0.696	0.997	0.266	0.990	0.252	0.999	0.661	0.996
	750	0.009	0.623	0.118	0.975	0.733	0.998	0.318	0.999	0.284	1.000	0.715	0.999
	1000	0.010	0.552	0.179	0.980	0.766	0.998	0.354	0.998	0.314	1.000	0.752	0.999
	1500	0.009	0.654	0.305	0.992	0.819	0.999	0.424	1.000	0.362	1.000	0.800	0.999
	3000	0.016	0.784	0.516	0.998	0.862	0.998	0.509	1.000	0.418	1.000	0.857	0.999

Table S7: Interval-based selection performance: mild saturation scenario. A patient is flagged if the lower bound of the 95% credible interval exceeds the threshold. Precision is averaged only over replicates in which the model flagged at least one patient, so it should be read together with sensitivity.

$\mu(x)$ Form	N	**BCF**						**SP-BART**					
		Top 10%		Top 25%		Top 50%		Top 10%		Top 25%		Top 50%	
		Sens	Prec	Sens	Prec	Sens	Prec	Sens	Prec	Sens	Prec	Sens	Prec
Linear	250	0.045	0.648	0.049	0.790	0.071	0.893	0.064	0.615	0.080	0.719	0.129	0.865
	500	0.052	0.880	0.061	0.962	0.109	0.972	0.071	0.823	0.105	0.920	0.192	0.967
	750	0.064	0.883	0.075	0.955	0.136	0.984	0.098	0.897	0.142	0.953	0.252	0.982
	1000	0.067	0.921	0.090	0.962	0.163	0.986	0.088	0.957	0.157	0.979	0.309	0.986
	1500	0.066	0.915	0.110	0.973	0.233	0.992	0.112	0.980	0.209	0.986	0.392	0.992
	3000	0.090	0.960	0.207	0.987	0.425	0.996	0.234	0.991	0.366	0.994	0.568	0.995
Non-linear	250	0.010	0.230	0.018	0.492	0.039	0.765	0.054	0.568	0.064	0.752	0.108	0.902
	500	0.012	0.298	0.021	0.617	0.068	0.869	0.068	0.832	0.107	0.927	0.186	0.972
	750	0.005	0.411	0.023	0.763	0.086	0.935	0.081	0.943	0.125	0.977	0.232	0.990
	1000	0.009	0.445	0.032	0.835	0.145	0.963	0.089	0.965	0.154	0.986	0.294	0.994
	1500	0.016	0.590	0.066	0.906	0.235	0.984	0.144	0.984	0.235	0.991	0.400	0.996
	3000	0.035	0.762	0.159	0.982	0.439	0.996	0.243	0.994	0.371	0.997	0.555	0.998

S6 Computational Details

We use Bayesian backfitting to alternate between the prognostic and treatment functions. Let u_i denote the fixed treatment coding, with $u_i = z_i$ for 0/1 coding and $u_i = z_i - 1/2$ for symmetric coding. Let h_i contain the p covariate columns and the q included interaction columns, and collect their coefficients in b. The treatment-weighted design W has rows $u_i h_i^\top$, so that $y = \mu + u\alpha + Wb + \varepsilon$ and $\tau(x_i) = \alpha + h_i^\top b$. Write $\mathcal{P}$ for the included interaction pairs, $K = p + q$, and

$$D = \operatorname{diag}\left(\lambda_1^2, \ldots, \lambda_p^2, \{\lambda_{j,k}^2 \lambda_{\text{glob}}^2\}_{(j,k)\in\mathcal{P}}\right), \qquad Q = W^\top W + D^{-1}.$$

All quantities below are on the fitted scale, after centering the outcome and dividing it by its training-sample standard deviation. The simulations use the generated covariates, while IHDP and CANVAS standardize continuous covariates before forming interaction products. Binary and categorical contrast columns and the treatment coding are not standardized. We use the inverse-gamma parameterization $p(v) \propto v^{-a-1}\exp(-b/v)$ for $v \sim \text{IG}(a, b)$; conditioning on the data and all other current parameters is denoted by $\mid -$.

For the prognostic function, we use the BART prior of Chipman et al. (2010), with $T = 250$ trees and splitting probability $0.95(1+d)^{-2}$ at depth d, a maximum depth of 10, and at least five observations per leaf. Given leaf variance s_μ^2, the leaf parameters have independent $\mathcal{N}(0, s_\mu^2)$ priors. For a leaf containing $n_{t\ell}$ observations, let $S_{t\ell}$ be the sum of the tree-specific partial residuals defined in Algorithm S1. Then

$$v_{t\ell} = \left(\frac{n_{t\ell}}{\sigma^2} + \frac{1}{s_\mu^2}\right)^{-1}, \qquad \theta_{t\ell} \mid - \sim \mathcal{N}\left(\frac{v_{t\ell} S_{t\ell}}{\sigma^2}, v_{t\ell}\right). \tag{S1}$$

With $s_\mu^2 \sim \mathrm{IG}(a_\mu/2, b_\mu/2)$ and L leaves across the forest, the leaf variance is drawn from

$$s_\mu^2 \mid - \sim \mathrm{IG}\left(\frac{a_\mu + L}{2}, \frac{b_\mu + \sum_{t,\ell} \theta_{t\ell}^2}{2}\right), \qquad a_\mu = 3, \quad b_\mu = 1/T. \tag{S2}$$

The treatment intercept and coefficients have the Gaussian full conditionals

$$v_\alpha = \left(\frac{u^\top u}{\sigma^2} + \frac{1}{100}\right)^{-1}, \qquad \alpha \mid - \sim \mathcal{N}\left(\frac{v_\alpha u^\top (y - \mu - Wb)}{\sigma^2}, v_\alpha\right), \tag{S3}$$

$$b \mid - \sim \mathcal{N}_K\left(Q^{-1} W^\top (y - \mu - u\alpha),\ \sigma^2 Q^{-1}\right). \tag{S4}$$

Following Makalic and Schmidt (2016), a unit half-Cauchy scale is represented by $\lambda^2 \mid \nu \sim \mathrm{IG}(1/2, 1/\nu)$ and $\nu \sim \mathrm{IG}(1/2, 1)$. Applying this to each local scale and using auxiliary variable ξ for the global scale gives

$$\nu_j \mid - \sim \mathrm{IG}(1, 1 + \lambda_j^{-2}), \qquad \lambda_j^2 \mid - \sim \mathrm{IG}\left(1, \frac{1}{\nu_j} + \frac{\beta_j^2}{2\sigma^2}\right), \tag{S5}$$

$$\nu_{j,k} \mid - \sim \mathrm{IG}(1, 1 + \lambda_{j,k}^{-2}), \qquad \lambda_{j,k}^2 \mid - \sim \mathrm{IG}\left(1, \frac{1}{\nu_{j,k}} + \frac{\beta_{j,k}^2}{2\sigma^2 \lambda_{\mathrm{glob}}^2}\right), \tag{S6}$$

$$\xi \mid - \sim \mathrm{IG}(1, 1 + \lambda_{\mathrm{glob}}^{-2}), \qquad \lambda_{\mathrm{glob}}^2 \mid - \sim \mathrm{IG}\left(\frac{q+1}{2}, \frac{1}{\xi} + \frac{1}{2\sigma^2} \sum_{(j,k) \in \mathcal{P}} \frac{\beta_{j,k}^2}{\lambda_{j,k}^2}\right). \tag{S7}$$

The main effects have local scales only; the global scale applies to the interaction block. Finally, with total residuals $e = y - \mu - u\alpha - Wb$,

$$\sigma^2 \mid - \sim \mathrm{IG}\left(1 + \frac{N + K}{2},\ 0.001 + \frac{e^\top e + b^\top D^{-1} b}{2}\right). \tag{S8}$$

The intercept and leaf priors are independent of σ^2; the regularized coefficient priors contribute the additional $K/2$ and $b^\top D^{-1} b/2$ terms.

The latest available parameter values are used at each step.

Algorithm S1 Bayesian Backfitting MCMC Scheme for SP-BART

Initialize: Prognostic forest $\mu(\cdot)$, leaf variance s_μ^2, linear parameters α, b, local and global shrinkage parameters, auxiliary variables, and noise variance σ^2.
Fix the covariates, interactions, treatment coding u, and propensity score (fitted in observational data or known from the randomized design).
for iteration $m = 1, \ldots, N_{\text{burn}} + N_{\text{keep}}$ **do**
    1. Update Prognostic Function (μ):
    Compute partial residuals for $i = 1, \ldots, N$:
        $R_i^* = y_i - u_i \tau^{(m-1)}(x_i)$.
    for each tree $t = 1, \ldots, T$ **do**
        Compute tree-specific residuals $R^*_{i,(-t)} = R_i^* - \sum_{s \neq t} g_s(x_i, \hat{\pi}_i)$.
        Propose a tree modification (e.g., grow, prune) and accept/reject via Metropolis–Hastings, integrating out leaf values.
        Draw the leaf parameters conditional on the updated tree structure and $R^*_{i,(-t)}$ using (S1).
    end for
    Draw the common leaf variance using (S2).
    2. Update Treatment Function (τ):
    Compute partial residuals for $i = 1, \ldots, N$:
        $R_i^{**} = y_i - \mu^{(m)}(x_i, \hat{\pi}_i)$.
    Draw the intercept α using (S3).
    Draw the main-effect and interaction coefficients jointly in b using (S4), with residuals $R^{**} - u\alpha$ and design W.
    Draw each local auxiliary and then its local scale using (S5) and (S6).
    Draw the global auxiliary ξ and then λ^2_{glob} using (S7).
    Update the CATE function:
        $\tau^{(m)}(x_i) = \alpha + \sum_{j=1}^{p} \beta_j x_{i,j} + \sum_{(j,k) \in \mathcal{P}} \beta_{j,k} x_{i,j} x_{i,k}$.
    3. Update Noise Variance (σ^2):
    Compute total residuals:
        $e_i = y_i - \mu^{(m)}(x_i, \hat{\pi}_i) - u_i \tau^{(m)}(x_i)$.
    Draw $\sigma^{2(m)}$ using (S8), including the coefficient-prior term $b^\top D^{-1} b$.
    if $m > N_{\text{burn}}$ **then**
        Store the draw and the predictions of interest.
    end if
end for

References

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